[Aditi Rao]

[TE EXTC]

Company: Russell Investments (RIIV India)

Job Profile: Application Developer Intern

Offer Type: Internship Only

Internship Stipend: INR 60000/-

CTC: Undisclosed

Location: Oberoi Garden City, Goregaon, Mumbai

Process Date:

- 15 Feb 2025: Application form released.
- 22 Feb 2025: Registration deadline.
- 4–5 Mar 2025: Shortlist calls sent out.
- 12 & 13 Mar 2025: Interview dates offered (12th March for most of the SPIT folk).
- 18 Mar 2025: Results declared.

Resume (at the time of application): Software___Aditi_Rao_RussellInvestments.pdf

(My Resume template is super mid. Please use Jake's Resume on Overleaf with the default settings, it's what I'm using RN)

Placement Process:

The process had a resume shortlisting round, after which we had 3 rounds at their office in Goregaon.

Round 1: Written Test (MCQs, and some Coding Questions)
Round 2: Technical Interview I
Round 3: Interview With the Director (Techno HR, with a focus on Tech)

1. **<u>Resume Shortlisting</u>** [Not really sure about the number for App Dev specifically, but across about 5-6 intern roles they had 9k applicants]

The application and shortlisting were entirely run by Russell Investments on Workday. SPIT's placement office circulated a short form so they could see who had applied, but the company did the actual screening, communication and interviews. HR communicated with students directly.

I remember receiving a call from HR around 5 o'clock somewhere around the end of February. Keep in mind to be polite and courteous during all such calls. That means, you must:

- Wait for the person speaking to complete what they've said.
- Slowly and carefully articulate any questions (and I mean, ANY) you may have. Be diplomatic in their framing.

For your reference, things that may have influenced their shortlisting:

a. My LinkedIn
b. My GitHub

Not linking my Leet/CF because it wasn't there in the resume I shared with Russell.

2. **Written Test (40 mins)** [87 candidates were shortlisted for this round] [NON ELIMINATORY]

We were invited to Russell's office at Oberoi for this round. It was a pen paper test, with 10 Technical MCQs on CS Fundamentals (DBMS, CCN, OS, OOPS, Data Structures), 10 Aptitude MCQs, and 3 DSA problems out of which we were expected to solve 2. The first problem expected the implementation of the Bubble Sort Algorithm. The second was to Find the duplicate in an array in O(n) and by using O(1) extra space, and the third was to check if a given string is valid, given certain heuristics (Valid Parentheses problem) [Use a stack]. These were not very complex problems. (I'm not sure how much it really mattered, but a lot of us (including me) **attempted all three coding problems**. Make of that what you will)

For this round, I encourage you to keep a calm and collected mind, and to practice writing code by hand. Do not, and I repeat, do **not** use ChatGPT while writing code in your primary language as much as you can avoid it.

3. **Technical Interview 1** [All candidates who wrote the test]

We were slowly called for an interview with a panel of 2 extremely experienced professionals. My interview was actually pretty late, but it extended so long I remember some of my friends scolding me for taking so long.

My interviewers were of varied domains. One was a highly experienced app developer, and the other was a DevOps engineer. This is important context to understand the rest of the interview

I walked in **(asked their permission before entering), smiled, and extended my hand to them to shake**. Start with the person closest to you, it's more natural. This gesture automatically familiarises you to them (especially since in tech, handshakes aren't all that common, so it helps break the ice).

We moved onto the interview.

They asked me to introduce myself. By then, we'd already had a short laugh over the water glass. This made the interview start with all of us smiling. I had a 3-min elevator pitch (fancy term for your answer to 'tell me about yourself') (I've memorized this and timed myself to stick to 3 minutes)

Then, we had the technical knowledge part of the interview first.

DSA:

1. Have you studied DSA? *Yes, I have*. What data structures comfortable with? *I am comfortable with Arrays, Linked Lists, Stacks, and Queues. While I do know and have studied Graphs and Trees, I cannot claim mastery*.
2. How would you **reverse this array in-place**?
3. They had my sheet. They asked me to **walk them through my implementations for all 3 codes**.
4. Suppose you're building a news aggregation app. A user searches for "climate policy India". Your backend search engine finds matching articles. But instead of just showing the title + link, you want to show a snippet of the article text that contains all three search terms, so the user immediately sees the context.
   To do that, you need to find the smallest window in the article text that contains all the query words. This is a **classic sliding window + hash map/frequency counter problem**.
   [Note: this is not the exact question, I have forgotten the exact one. But I remember the implementation, so I have done my best to create a similar problem statement so you can relate to the scenario]

DBMS:

1. A JOIN Query (joining 3 tables)
2. Do you know what a view is?
3. What's an index?
4. Find the Top-N Highest Salary

CCN:

1. TCP vs UDP
2. What happens when you click a URL?
3. What's a REST API?

Now, here's a place where my resume got me into a spot of trouble.

I have mentioned I know ‘RESTful APIs’ in my skills. I do know REST APIs. Thing is, I didn’t know what the ‘-ful’ part meant. (Arre REST, RESTful, same hi laga tha) So my interviewer asks me, **this is a REST API. You’ve mentioned a *RESTful* API. What’s that?**

I didn’t know at all. So, I smiled and said, I know how REST APIs work and I’ve built them, but I’m not fully sure what the ‘-ful’ part means. I assume it has something to do with the principle of REST → statelessness, resource-based URIs, use of HTTP verbs, etc. [PS: I made this up. I didn’t know at the time either. By sheer luck, this is the right answer. So, the moral of the story, *say something*]

Resume:

1. I had an Embedded systems experience on my resume. Interviewers are usually more than happy to hear about your experience. Talk about it with as much passion as you can muster. You must be able to explain what the company does, how you contributed, so on and so forth. If you were at a **startup**, highlight how you **wore multiple hats, learned quickly, and adapted to a fast-moving environment.** If it was a **larger company, talk about working within established systems, collaborating across teams, and how your work contributed to the bigger picture**. Personally, I talked to them about the product of the startup, how my work was on the product itself, and how I fixed a critical bug in the firmware code.
2. One of my interviewers was an experienced App Developer. He was extremely interested in the BLE (Bluetooth Low Energy) work I’d done at my internship. Here’s what he asked me:
   -> How does BLE connection establishment normally work?
   -> Why did you need a custom three-way handshake instead of relying on BLE’s built-in connection mechanisms?
   -> How did you measure the “74% reduction” in sync issues?

None of this may apply to you. But the questions were pretty in-depth (BLE is just one line on my resume) so be prepared for this kind of deep-dive

3. The other interviewer, a DevOps engineer, was very interested in my project Percy, an AI-driven system optimizing CI/CD pipelines, automating testing, enhancing the developer experience. She asked me for the **system design**, which I drew on a sheet with simple boxes and arrows. She had many questions about my design choices (**Why are you maintaining the data in a JSON, why are you using Celery here, how would you integrate this into an existing CI system**) She asked me what an **AST was, and how it helped in documentation generation.**
   She also interspersed some technical knowledge questions, like what’s the **difference**

between on-prem and cloud servers, what you'd have to take care of in an on-prem server which cloud providers cover for you (I probably didn't answer this as well as I could have)

Once they were done with their questions, they asked me if I had any questions for them. Honestly, I did (like have actual questions. Not just 'ask for the sake of asking, ask something')

1. What does the work of an App Developer look like at Russell?
2. Do you attend client meetings?
3. How are teams structured?
4. What are your office timings like?
5. How does your CI/CD pipeline work, could you please give me a very very quick overview?

4. **[Round 3 Name]** [14 candidates appeared for this round]

My round 2 was with a senior director who sits in London. He'd come to India for some other work. He was in charge of their data (very important in any financial firm) and had a very intimidating presence. I will admit, I was very terrified. It had been a long wait (this interview happened at 4:30, when my previous interview had ended at 1ish) so the tension had had time to build up as well.

He was very polite. I shook his hand with a smile. He asked me to work him through my resume (the same 3 min elevator pitch)

He was very, *very* interested in my experience. He wanted to know all about the product, the use, and the clinical trials associated with the product. He asked me about how a **day working at a startup as a firmware intern was, how I debugged the cause of the lag in transmission, how the fix was approved, and how we tested it.** He wanted to know about my knowledge of **Alpha/Beta testing, SDLC.**

Then, he asked me to write a simple SQL query to find all entries for October 2024, sorted by day, given that each entry was in DATE format in SQL. Now, this is an easy question, but I **blanked.** I cannot explain why. I could not remember **LIKE in SQL,** probably because I was so intimidated. In this situation, **drink water.** If you know something, but you're unable to answer, drink water. It provides a separate stimulus to your mind and helps clear stress.

Then, I said – the exact term is slipping my mind, but this is how the query would be structured. And as I wrote that down, I remembered LIKE and wrote that down as well.

He seemed happy with the answer. He moved onto some technical questions:

1. On-prem versus cloud (here, I ended up cross-questioning him a bit too. Keep it conversational! He seemed to want to talk)
2. How data is stored in a financial firm

Then, he shifted to cultural fit questions.

1. Tell me a situation where you were frustrated with a team mate. How did you work past the frustration?
2. What, according to you, is the most challenging part of working with a team?

The rest has slipped my mind, but it was somewhat around this.

Once he was done, he asked if I have any questions for him. I asked questions based on the talks we'd had in the morning with the Director of Application Development–

1. Could you tell me why you've opted for Snowflake as your Cloud Vendor?
2. How do firms like Russell deal with damages to on-prem servers?
3. What would you expect from an intern like me?

Useful Tips:

1. The most important advice for any interview I could give you is to **always smile.** Be graceful about your mistakes.
2. **Do not claim to know something you do not know.** That is a massive red flag.
3. The interviewers are human. Appeal to their human nature. They're tired and bored after long days of continuous interviews too.

Resources (For the lazy folk):

1. **SQL**: DataQuest for LMR, SQLZoo for a one sitting full syllabus study. Practice Leetcode SQL if the syntax tends to slip from your mind and if application of certain functions tends to slip from your mind.
2. **DSA**: The usual resources are great. Follow those diligently. If you have the time and you love coding, give the CSES Problemset a shot. I'm at the Mathematics section and it's really fun!
3. **OS**: Any YT course will do if you're studying it for the first time. If you're reading this the night before the interview, just read up on Processes and Threads, Process Scheduling, Synchronization, Memory Management, Filesystems and I/O. If you have more time, this Overview by InterviewBit is pretty swell!
4. **CCN**: Same as OS. LMR. Very, very LMR (Ye toh karke hi jao pls).

Mistakes that I made that I feel you can learn from:

1. I thought Aptitude was silly. It still is, but trust me, it's better to be able to do it faster than to have to walk yourself through every problem every time. That is tiring and gets old fast, and you often end up running out of time. **Practice Aptitude**. It never hurts.
2. **Keep your primary language your primary language.** That means that if you do DSA in C++, please know the ins-and-outs of the language. Do not go about mastering all languages. *If you're just starting out, it may be simpler for you to do your DSA practice in Java,* just to cover your bases with technical MCQs (Russell didn't care about language, they cared much more about the depth of your technical understanding, but not all companies are like this)
3. Write. Code. Feels dumb? Trust me, it isn't. That notebook is the OG IDE.

PS: Read all the interview experiences in the folder. Everyone has covered what they remember of the process, so everyone may have missed something or the other.

Placements need skill and practice. But they're also, to an extent, very much a game of LUCK. Do not be demotivated by rejections. Be confident in yourself.

OFF-CAMPUS IS VERY MUCH A THING, AND YOU MUST WORK FOR THAT TOO. All career migrations post-college will be off-campus! So prep for off-campus!

I'm always available on LinkedIn! You can also reach out to me on +91 8779410757 (Whatsapp)

Best wishes for your placements!